25900014 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBIF100 - BIOMOLECULES AND MOLECULAR VISUALIZATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Comment on peptide bond? [U] / [CO1]
2. Name any two types of RNAs found in living beings. [A] / [CO2]
3. What are the three major forms of DNA secondary structure? [K] / [CO2]
4. Chart the main types of biological databases. [A] / [CO3]
5. Operate a pairwise sequence alignment tool for comparing two sequences. [An] / [CO4]
6. Write one advantage of using Swiss PDB Viewer for molecular visualization. [A] / [CO5, CO6]
7. Discover which organization developed Map Viewer. [A] / [CO5, CO6]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

8. How would you explain the role of glycogen in physical activity? [A] / [CO1]

9. What are the three main types of RNA, and what are their roles? [U] / [CO2]

10. Explain the importance of MSA in identifying conserved protein domains. [An] / [CO4]

11. Interpret ribbon model and its applications. [A] / [CO5, CO6]

12. Explain molecular graphics and its types. [A] / [CO5, CO6]

[5x3 = 15]

Part C

Long essay Questions

Answer any **1** questions

Each question carries **10** marks

13. What is the difference between monounsaturated and polyunsaturated fatty acids? Give examples. [U] / [CO1]

14. Compare sequence databases and structure databases with examples. [A] / [CO3]

15. Explain MolScript, and what is its primary function in molecular visualization. [S] / [CO5, CO6]

[10x1 = 10]